DRAM Quadrants

Testing and evaluation strategy for 4164 / 4532 / 3732 RAM chips

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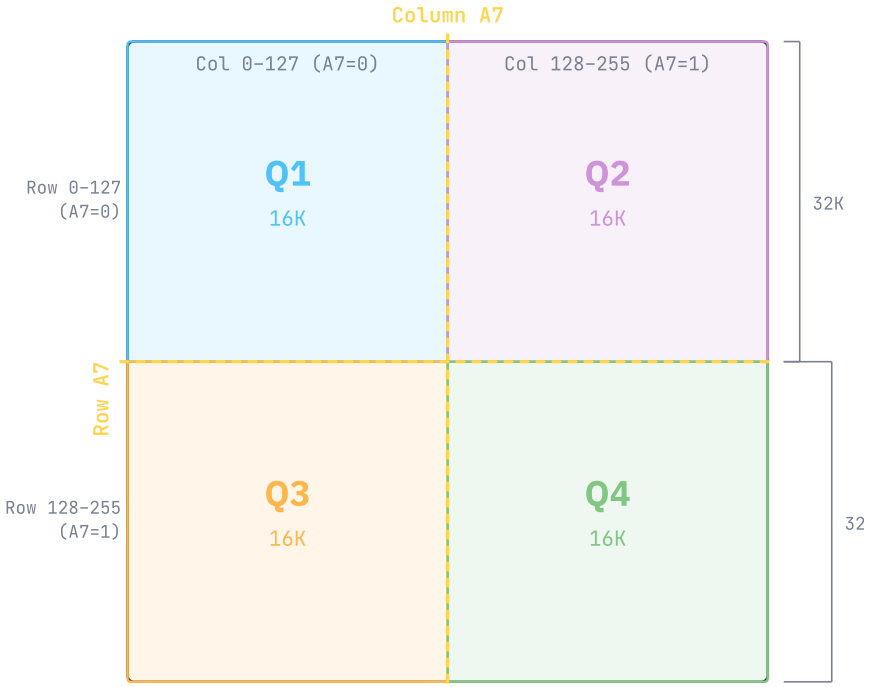
▸ What are the Quadrants?

A `4164` (64K×1) DRAM is internally organized as a matrix of **256 Rows × 256 Columns**. Addressing uses 8 address lines `A0–A7`, which select the row during RAS and the column during CAS.

If we split the matrix at the most significant address bit `A7` — both horizontally (rows) and vertically (columns) — we get **four equal quadrants** of 128×128 = 16,384 bits (16K) each:

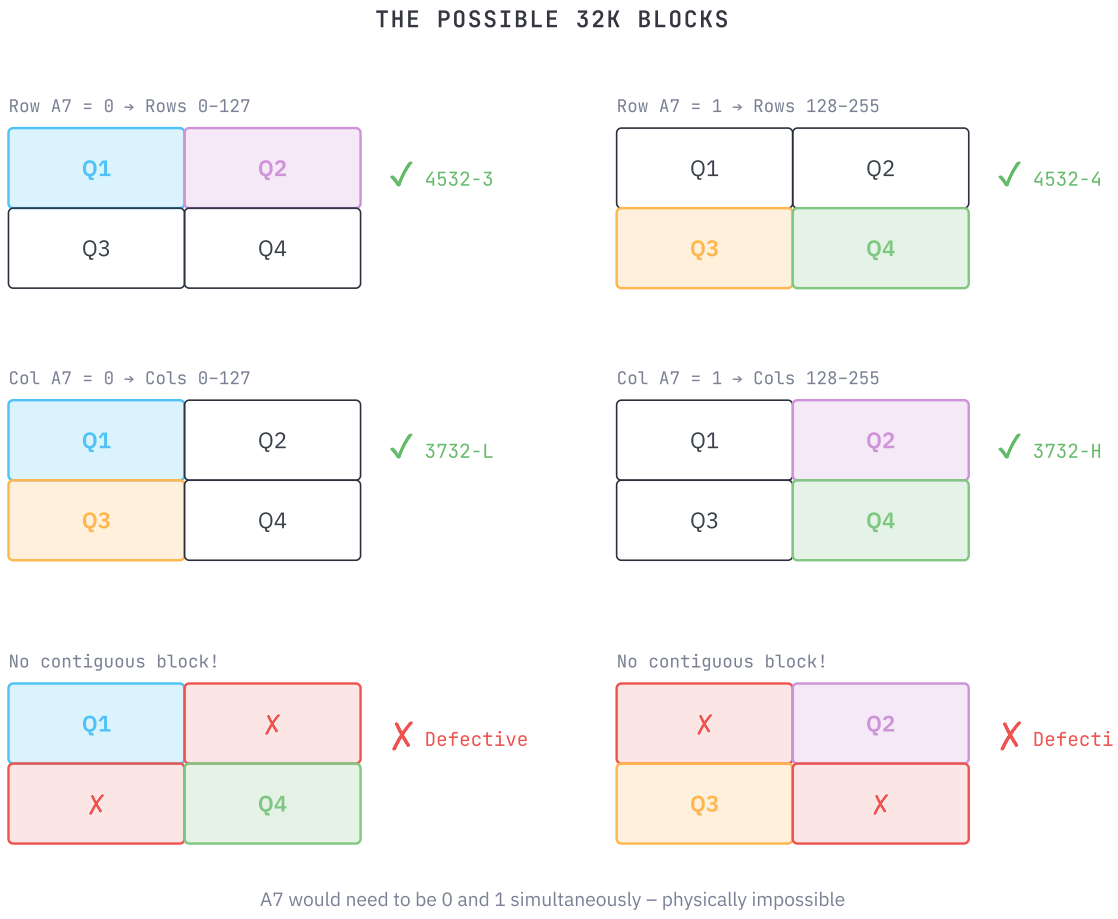
4164 - MATRIX SPLIT AT A7



> **Core idea:** A defective `4164` may still be usable as a `TMS4532` (32K×1) or `MSM3732` (32K×1) — provided that a contiguous 32K block is error-free.

▸ Valid 32K Blocks

A 32K×1 DRAM requires a **contiguous address space**. The `TMS4532` splits the matrix at `Row A7` — only an upper and a lower half exist. The `MSM3732` can additionally split at `Column A7`:



Chip Type Classification

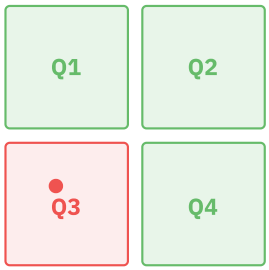
Depending on which quadrants are error-free, the chip's suitability as a 32K chip is determined:

Q1	Q2	Q3	Q4	Result	Explanation
✓	✓	✓	✓	4164	Fully functional 64K×1 chip
✓	✓	✗	✗	4532-3	Upper half (Q1+Q2) – Lower Rows OK
✗	✗	✓	✓	4532-4	Lower half (Q3+Q4) – Upper Rows OK
✓	✗	✓	✗	3732-L	Left half (Q1+Q3) – no 4532 possible (column split)
✗	✓	✗	✓	3732-H	Right half (Q2+Q4) – no 4532 possible (column split)
✓	✗	✗	✓	Defective	Diagonal – no contiguous 32K block possible
✗	✓	✓	✗	Defective	Diagonal – no contiguous 32K block possible

Practical Examples

The following diagrams show typical error patterns and their impact on chip classification. Red cells indicate defective areas.

Example 1: Single bit error in Q3

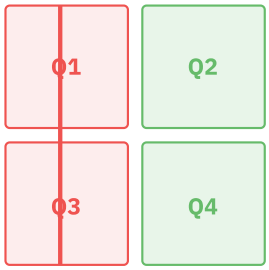


✓ 4532-3 (Q1+Q2) · ✓ 3732-H (Q2+Q4)

Example 2: Entire row defective (Row 200)

✓ 4532-3 (Q1+Q2) – only upper half usable

Example 3: Column defective (Col 50)



✓ 3732-H (Q2+Q4) – only right half usable

Example 4: Diagonal errors

✗ Defective – no contiguous 32K block

Example 5: All quadrants OK

✓ 4164 – full 64K×1 chip

Example 6: Multiple errors, but Q1+Q3 clear

✓ 3732-L (Q1+Q3) – left half usable

Why "diagonal" doesn't work

A 32K×1 DRAM has **7 instead of 8 address lines**. Compared to the 4164, one address line is missing – `A7` is hardwired to either **0** or **1**.

For the `TMS4532`, `A7` is fixed on the **row side** → upper or lower half.
For the `MSM3732`, `A7` is fixed on the **column side** → left or right half.

A **diagonal combination** (e.g. Q1+Q4) would require `Row-A7=0` with `Col-A7=0` *and* `Row-A7=1` with `Col-A7=1`. Regardless of whether row or column is fixed – a diagonal selection is never a contiguous address block. `A7` cannot be two values at once.

Chip Nomenclature

TMS4532 (Texas Instruments)

The TMS4532 is a tested 4164 where only one contiguous 32K block along the **row axis** is functional. The split is exclusively at `Row A7`:

Type	Usable Half	A7 Configuration
`TMS4532-3`	Upper half (Q1+Q2) – Lower Rows 0–127	Row A7 = 0
`TMS4532-4`	Lower half (Q3+Q4) – Upper Rows 128–255	Row A7 = 1

OKI MSM3732 (32K×1)

The MSM3732 uses the **column axis** for its split at `Column A7`:

Type	Usable Half	A7 Configuration
`MSM3732-L`	Left half (Q1+Q3) – Low Columns 0–127	Column A7 = 0
`MSM3732-H`	Right half (Q2+Q4) – High Columns 128–255	Column A7 = 1

Note: The TMS4532 only exists in variants -3 and -4 (row-based split). The MSM3732 uses a column-based split with -H and -L suffixes. A defective 4164 may qualify as either a 4532 *or* a 3732 depending on the error pattern – but not as both, since the split axis is different.

Evaluation in the RAM Tester

The RAM Tester performs the following steps:

1. Full 4164 test across all 256×256 addresses using multiple test patterns (0x00, 0xFF, 0xAA, 0x55, random) and a retention test.

2. Error localization: When errors occur, the row and column address is recorded. The affected quadrant is determined based on `A7` of the failing address.

3. Quadrant evaluation: Each of the four quadrants is checked for errors. This determines the possible 32K configurations.

4. Classification: The tester reports whether the chip qualifies as a 4164 (error-free), a 4532/3732 variant (one 32K block error-free), or as defective (no contiguous error-free block).